

Lecture 17: The spherical Earth

David Al-Attar

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Outline and Motivation

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Historical Context

The methods described are foundational to seismology. It's remarkable that much of our knowledge of Earth's deep structure was determined long before the advent of modern digital computers.

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- For this, we need an accurate model of the Earth's seismic velocity structure.
- **Ray theory** is used to efficiently calculate the travel time from a potential hypocentre to each seismometer.

The Inverse Problem for Location

The problem of finding the hypocentre is a classic inverse problem.

- The measured arrival time t_i at seismometer i is related to the source parameters by:

$$t_i = t_h + T(\mathbf{x}_i, \mathbf{x}_h) + e_i$$

where $T(\mathbf{x}_i, \mathbf{x}_h)$ is the ray-theoretic travel time and e_i is measurement error.

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Least-Squares Minimisation

We have four unknowns (t_h and the three components of $\mathbf{x}_h$). We estimate them by finding the values that minimise the sum of squared, weighted differences between observed and predicted arrival times:

$$J(\mathbf{x}_h, t_h) = \sum_{i=1}^{N_s} \frac{[t_h + T(\mathbf{x}_i, \mathbf{x}_h) - t_i]^2}{\sigma_i^2}$$

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- **Station Geometry:** An even distribution of seismometers in distance and azimuth around the source gives the best constraint.
- **Velocity Model Error:** Any inaccuracy in the assumed Earth structure will map directly into errors in the location.

Rays in a Spherical Earth: Setup

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- The ray path $\mathbf{x}(\sigma)$ and slowness $\mathbf{p}(\sigma)$ are given by:

$$\frac{d\mathbf{x}}{d\sigma} = \frac{\partial H}{\partial \mathbf{p}} \quad , \quad \frac{d\mathbf{p}}{d\sigma} = -\frac{\partial H}{\partial \mathbf{x}}$$

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Goal

Because of the spherical symmetry, this system of ODEs has a conserved quantity, which allows us to simplify the problem dramatically.

A Conserved Quantity: The Ray Parameter

Due to the spherical symmetry, the quantity $\mathbf{q} = \mathbf{x} \times \mathbf{p}$ is conserved along a ray.

$$\frac{d}{d\sigma}(\mathbf{x} \times \mathbf{p}) = \frac{d\mathbf{x}}{d\sigma} \times \mathbf{p} + \mathbf{x} \times \frac{d\mathbf{p}}{d\sigma} = 0$$

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- As a consequence, the ray path is always confined to the plane containing the Earth's center, the source, and the receiver.

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The Ray Parameter

The magnitude of this conserved vector, q , is known as the **ray parameter**. It is constant for any given ray.

$$q = \|\mathbf{x} \times \mathbf{p}\| = r\|\mathbf{p}\| \sin \theta = \frac{r \sin \theta}{\alpha(r)}$$

Here, θ is the angle between the position vector $\mathbf{x}$ and the ray direction $\mathbf{p}$.

Travel-Time Integrals

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Travel-Time and Distance Formulae

For a ray that turns at radius r_t and returns to the surface at radius a :

- **Total Distance:** $\Delta(q) = \int_{r_t}^a \frac{2\alpha(r)q}{r\sqrt{r^2 - \alpha(r)^2 q^2}} dr$
- **Total Time:** $T(q) = \int_{r_t}^a \frac{2r}{\alpha(r)\sqrt{r^2 - \alpha(r)^2 q^2}} dr$

Determining Earth's Velocity Structure

The Inverse Problem

By measuring the travel times (T) of many seismic phases at various epicentral distances (Δ) from thousands of earthquakes, seismologists have built up global travel-time curves. The inverse problem is to find the velocity structure $\alpha(r)$ that best explains these curves.

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Key Historical Discoveries

This approach led to the discovery of Earth's major layers:

- **Richard Oldham** first identified the core.
- **Harold Jeffreys** argued for a fluid outer core.
- **Inge Lehmann** discovered the solid inner core.

Overview of Earth's Velocity Structure (PREM)

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- The general trend is that velocity increases with depth due to increasing pressure and density.
- This smooth trend is interrupted by several sharp discontinuities.

(A graph showing P-wave and S-wave velocity vs. depth would go here)

Interpreting the Discontinuities

The major seismic discontinuities mark changes in composition or phase.

- **Core-Mantle Boundary (CMB):** A major compositional change from silicate rock (mantle) to liquid iron-nickel alloy (outer core). The fact that the outer core is liquid is confirmed by the inability of S-waves (which require shear strength) to travel through it.

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- **Inner-Core Boundary (ICB):** A phase change where the iron alloy freezes due to immense pressure, forming the solid inner core.
- **Mantle Transition Zone (410 & 670 km):** These are not compositional boundaries. They are solid-state phase transformations where mantle minerals rearrange into denser crystal structures under increasing pressure.

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- How global travel-time curves are used as data in an inverse problem to determine the Earth's spherically averaged velocity structure.
- The basic features of Earth's velocity structure: the crust, mantle, liquid outer core, and solid inner core, and the physical interpretation of the major discontinuities (CMB, ICB, 410, 670).